Course Code	Course Name	Theory	Practical	Tutorial	Theory	Oral	Tutorial	Total
IoTCSBCDO 7011	**Advanced Cloud Computing Security**	03	--	--	03	--	--	03

Course Code	Course Name	Examination Scheme							
		Theory Marks				Term Work	Practical	Oral	Total
		Internal assessment			End Sem. Exam				
		Test1	Test 2	Avg. of 2 Tests					
IoTCSBCD O7011	**Advanced Cloud Computing Security**	20	20	20	80	--	--	--	100

Course Objectives:

Sr. No.	Course Objectives
The course aims:	
1	To understand the concept of security and its significance in the context of cloud computing.
2	To study cloud infrastructure security and mitigation techniques
3	To understand the working of Data center and Data Protection techniques
4	To develop a comprehensive understanding of challenges and solutions in secure identity management for cloud environments
5	To study Compliance and Security Audits policies for cloud data
6	To understand the Cloud Native Security

Course Outcomes:

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
On successful completion, of course, learner/student will be able to:		
1	Understand the concept of security and its importance in the context of cloud computing.	L2
2	Analyze cloud infrastructure security and apply different mitigation techniques.	L3, L4
3	Apply different data protection techniques in data centers.	L3
4	Design and implement secure identity management solutions for cloud environments	L6
5	Interpret and appropriately apply the policies on Compliance and Security Audits for cloud data	L2, L3
6	Demonstrate cloud security tools for designing, implementing, and managing cloud-native security	L2, L6

Prerequisite: Knowledge of Cloud Computing and Cryptography and Network Security

DETAILED SYLLABUS

Sr. No.	Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Basics of cloud computing, network and system security	**2**	
I	**Fundamentals Of Cloud Security Concepts**	What is security, why is it required in cloud computing, Different types of security in cloud, attacks, and vulnerabilities **Cloud Security Concepts** - CIA Triad (Confidentiality, integrity, availability), privacy, authentication, non-repudiation, access control, defence in depth, least privilege, Traditional vs Cloud Security, importance, challenges in different cloud environment (public, private, hybrid, muti-cloud) **Self-Learning Topic:** Real-world Example of CIA Triad - Bank ATM	**5**	**CO1**
II	**Cloud Infrastructure Security: Threats and Mitigation Techniques**	**Secure Infrastructure architecture** **Infrastructure Security:** Network Level, Host Level and Application Level Common attack vectors and threats **Mitigation techniques**- Isolation, Virtualization and Segmentation, Intruder Detection and prevention, Firewall, OS Hardening and minimization, Verified and measured boot. **Self-Learning Topics:** DoS, Man-in-the-Cloud, Insecure APIs, Insider Threats, Cookie Poisoning, Cloud Malware Injection,	**7**	**CO2**
III	**Cloud Data Security**	Cloud security principles Aspects of Data Security **Mitigation techniques**: Data retention, deletion and archiving procedures for tenant data, Encryption, Data Redaction, Tokenization, Obfuscation, PKI and Key **Data center Security and Data Protection:** Physical and network **data** center security, Implementation of security in Virtual Data centers, East-west Traffic Protections, Types of firewall, IDS and IPS, DMZ **Provider Data and Its Security** **Self-Learning Topics:** **Case studies:** Capital One Data Breach, Uber's AWS Data Breach, Dow Jones Data Leak, Accenture AWS S3 Data Exposure, Verizon AWS S3 Data Exposure	**6**	**CO3**
IV	**Secure Identity Management in The Cloud: Challenges And Solutions**	IAM overview, Trust Boundaries and IAM, Architecture / Lifecycle process, IAM standards and protocols, IAM Challenges **Cloud Authorization Management:** **Identity management** - User Identification, Authentication and Authorization **Roles-based Access Control** - Multi-factor authentication, Single Sign-on, Identity Federation **Cloud Service Provider IAM Practice** **Self-Learning Topic:** IAM service in AWS	**6**	**CO4**
V	**Disaster Recovery Auditing: Mitigating Risk and Ensuring Compliance**	Cloud disaster recovery, types of disasters recovery, benefits of disaster recovery, cloud disaster recovery planning **Privacy:** Data life cycle, key privacy concerns in cloud, privacy risk management and compliance, legal and regulatory implications, **Cloud Audit and Compliance:** Internal Policy Compliance, Governance, Risk, and Compliance (GRC), Benefits, GRC Program Implementation, Cloud Security Alliance, **Self-Learning Topics:** HIPAA, ISO, PCI	**7**	**CO5**

VI	**Cloud Native Security in The Modern Organization**	Overview of Cloud Native Security, where it fits in the Modern Organization, purpose of Security, Cloud Native Security Architecture, Threats to Cloud Native Applications 3 R's and 4 C's of Cloud Native Security Cloud Native Security Controls, Cloud Native Security Tools, Cloud Native security architecture principles, DevSecOps, How to Measure the Impact of Security, Cloud-Native Application Protection Platform (CNAPP) **Self Learning Topic:** Case study on Secure the Cloud	**6**	**CO6**

Textbooks:

1. Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance by Tim Mather, Subra Kumaraswamy, and Shahed Latif, O'Reilly
2. Cloud Native Security Cookbook: Recipes for a Secure Cloud 1st Edition by Josh Armitage , O'Reilly
3. Cloud Security: A Comprehensive Guide to Secure Cloud Computing by Ronald L. Krutz and Russell Dean Vines, Wiley

References:

1. "Securing the Cloud: Cloud Computer Security Techniques and Tactics" by Vic (J.R.) Winkler, SYNGRESS
2. "Identity and Access Management as a Service: Security as a Service" by Wei Meng Lee
3. Cloud Security for Dummies by Ted Coombs, O'Reilly

Online References:

1. https://www.coursera.org/learn/cloud-computing-security#about
2. https://www.coursera.org/specializations/cybersecurity-cloud
3. https://www.edx.org/course/cloud-computing-security
4. **https://www.ibm.com/topics/cloud-security**
5. **https://www.vmware.com/topics/glossary/content/east-west-security.html**
6. **https://www.vmware.com/topics/glossary/content/data-center-security.html**
7. **https://cloud.google.com/learn/what-is-disaster-recovery**
8. **https://www.splunk.com/en_us/blog/learn/cloud-native-security.html**

Assessment:

Internal Assessment (IA) for 20 marks:

● IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test

Question paper format

● Question Paper will comprise of a total of **six questions each carrying 20 marks.** Q.1 will be **compulsory** and should **cover maximum contents of the syllabus.**

● **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)

● A total of **four questions** needs to be answered.